

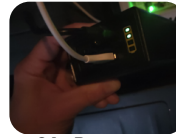
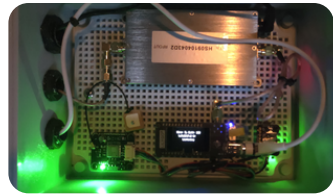


Blackout Comms High Power Stealth Link - DIY

On the Outside
Stealth Antenna



Storage Compartment or Under Seat



24v Battery
5-6 Days / Charge

- Store / forward packets & location
- GPS tracker (if GPS installed)
- Conceal in a storage area
- Amp Options: None, 1 watt, 20 watt

Required Components

One of the Following Time or GPS Sources, along with a Qwiic wire to connect it:

- [Adafruit DS3231 RTC](#)
- [SparkFun RV-8803 RTC](#)
- [DFRobot 1103 GNSS](#)
- [DFRobot TEL0157 GNSS](#)
- [SparkFun SAM-M8Q GNSS](#)

Amp Options (Not Required)

1 Watt

- [Sunhans 1W Amp](#)
- [Talentcell 12v](#)

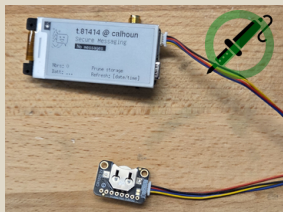
20 Watts

Amateur radio
license required!

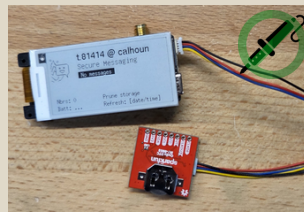
- [Szuashi 20W Amp](#)
- [Talentcell 24v](#)

1. Choose one of the following 7 link options:

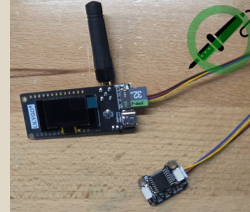
Time Node Options - Less power consumption than GPS links - *No Soldering*



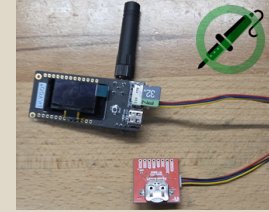
T3S3 Paper + Adafruit DS3231



T3S3 Paper + SparkFun RV8803



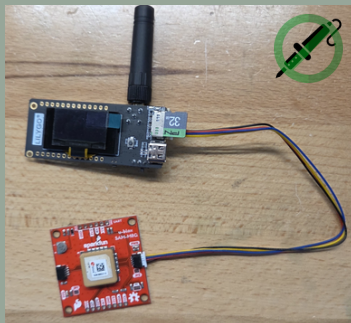
T3S3 + Adafruit DS3231



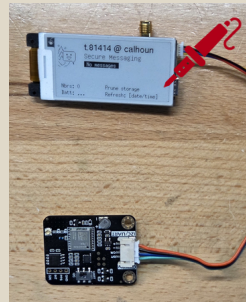
T3S3 + SparkFun RV8803

Connect a supported RTC to the T3S3 using a standard Qwiic/Stemma QT cable

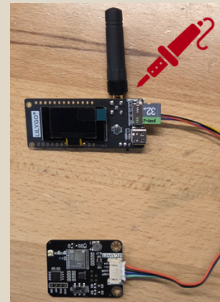
GPS Options - Use more power, but securely broadcast location regularly



T3S3 + SparkFun GNSS (uBlox)



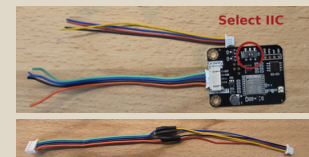
T3S3 Paper + DFRobot GNSS



T3S3 + DFRobot GNSS

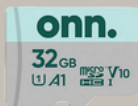
If you choose a DFRobot option, you'll need to create a Qwiic adapter:

- DFR Green → Qwiic Blue
- DFR Blue → Qwiic Yellow
- DFR Red → Qwiic Red
- DFR Black → Qwiic Black



2. Add Compatible SD Card

See [firmware download page](#) for list



3. Install Firmware

<https://chatters.io/firmware>

4. Onboard the Link

- Power up the link
- On your Root communicator, select "Settings / Cluster / Onboard New Device"
- Wait a minute or two, it should automatically join



Additional Components (optional)

You may, at any time, add a compatible relay and/or radar motion sensor, and the link will automatically begin to utilize it. See chatters.io/support for a list of compatible modules.



- Power the node off
- Add the sensor/module into the Qwiic chain
- Power the link back on

Within a few minutes, nearby devices should begin to recognize the motion sensor or whatever you added.

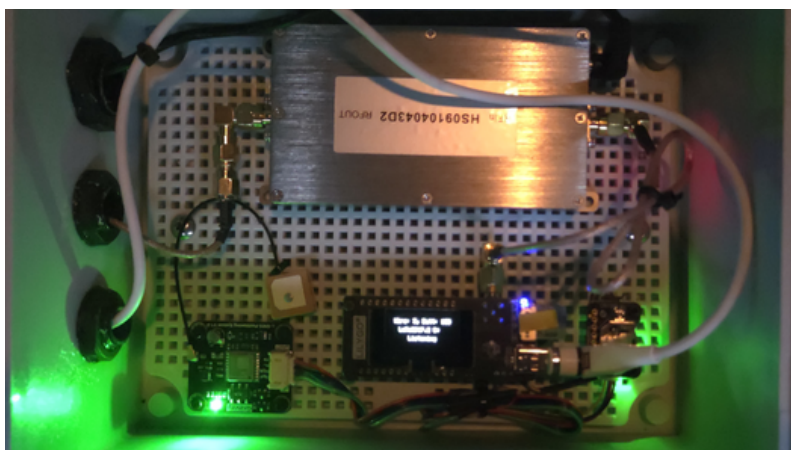
Create Openings in the Enclosure



You'll need to figure out how to do this, based on the enclosure you have selected. I made openings for:

- 12/24v power cable
- 5v power cable
- Antenna / Coax Cable

Install Components into Enclosure



I have chosen the 20 watt amp with T3S3 OLED, Adafruit RTC, and DFR GPS. The battery will be outside the enclosure, so there is plenty of room, as you can see.

Install the Antenna



I can't provide detailed instructions here, as this will vary depending on where your mounting this antenna and what type of antenna you choose.

I chose the Laird Phantom Stealth antenna mounted magnetically, so links for that are below.

[Laird Phantom Stealth Antenna](#)
[NMO Magnetic Mount](#)



You will also need a few SMA and other RF connectors/cables, which will vary, depending on the components you've selected.